7.6 Answers: ratio tables

← Unit 7.6 · Practice

Many routes are possible. The steps given here are one sensible route; any correct route deserves a tick.

Set 1 · Warm-up: the moves

1. 12 (× 2)
2. 24 (× 2 again)
3. 60 (× 10 from 1 pack)
4. € 12 (× 3)
5. € 20 (× 5)
6. € 6 (× 2)
7. € 1,50 (: 2 from 2 kg)
8. € 0,80 (: 10)
9. € 4 (half of 10 pens)
10. 10 wheels

Set 2 · Building: recipes

1. 500 g (× 2)
2. 1 000 ml (= 1 litre)
3. 1 egg (: 2)
4. 250 ml (: 2)
5. 375 g (4 people + 2 people: 250 + 125)
6. 750 ml (500 + 250)
7. 6 eggs (× 3)
8. 625 g (8 people + 2 people: 500 + 125)
9. 125 ml (4 people : 4)
10. 2 eggs

Set 3 · Building: prices

1. € 4,80 (× 2)
2. € 0,80 (: 3)
3. € 7,20 (× 3)
4. € 12 (× 2)
5. € 3 (: 2)
6. € 9 (500 g + 250 g: € 6 + € 3)
7. € 4,20 (3 × € 1,20 = € 3,60, plus half of € 1,20 = € 0,60)
8. The 400 g bag (€ 0,80 per 100 g, against € 0,90 for the 250 g bag)
9. € 2,70 (€ 1,80 + € 0,90)
10. € 4

Set 4 · Building: speed

1. 30 km
2. 7,5 km
3. 45 km
4. 22,5 km (15 + 7,5)
5. 4 hours (4 × 15 = 60)
6. 3 km (half an hour)
7. 1 km (10 minutes is a sixth of an hour; 6 : 6)

8. 30 km (15 minutes is a quarter of an hour; 120 : 4)
9. 45 minutes (30 km takes 15 minutes, so 90 km takes 3 × 15)
10. 200 km

Talk about it: 60 divides nicely by 30, 15 and 10, but not by 25.

Set 5 · Stretch: mixed problems

1. 2,5 litres (8 m² → 1 l, 16 m² → 2 l, 4 m² → 0,5 l)
2. 7,5 litres (150 : 20)
3. 200 ml (6 parts make 1 200 ml, so 1 part is 200 ml)
4. 450 g (6 → 3 people is 150 g; 6 + 3 = 9 people is 300 + 150)
5. 5 minutes (8 pages per minute; 40 : 8)
6. € 52,50 (1 ticket = € 7,50; × 7)
7. Ruben (Ruben: 18 km in 60 minutes; Sara: 16 km in 60 minutes)
8. € 3

Puzzle corner

Mix A is stronger. In Mix A, 2 of the 5 cups are syrup (2/5 = 16/40); in Mix B, 3 of the 8 cups are syrup (3/8 = 15/40). Close, so tasting may not settle it, which is why the maths is useful. Mix C is a genuinely hard question; trying is the point. A mix *somewhere in between* is easy to find: pour Mix A and Mix B together (5 cups syrup, 8 cups water, so 5/13 syrup). *Exactly* halfway is 31/80 syrup (halfway between 32/80 and 30/80), so 31 cups of syrup with 49 cups of water: a very big jug.